

[Continuation of Paper 147 — A Memoir on the Symmetric Functions of the Roots of an Equation, from PDF p. 451 = print p. 435.]

[Print pp. 435–439 (PDF pp. 451–455) consist almost entirely of the large numerical Tables of the symmetric functions for partitions of weights ≤ 10 . These are the continuation of Table V (10) introduced in earlier pages of the paper, and Table X (6) (which “Runs on to p. 439”). Each table is a wide cross-tabulation: the rows are indexed by partitions $(\alpha_1\alpha_2\dots)$ (such as (10), (91), (82), (73), (64), (5^2) , (81^2) , (721), (631), (541), (62^2) , (532), (4^22) , (43^2) , (71^3) , (621^2) , (531^2) , (4^21^2) , (52^21) , ...) while the columns are indexed by monomials in the coefficients $a, b, c, d, e, f, \dots$ such as b^4d^2 , b^3c^3 , bcd^3 , $\bar{a}c^2d^2$ etc. (with the convention $a = 1$ in the body of the table). The entries are signed integers giving the coefficient with which each monomial appears in the symmetric function corresponding to a given row partition. The bottom line of each table gives the column-sum (the numerical verification of the row of coefficients). The detailed tables run for several full pages and are not reproduced here in numerical form: their structure follows that of the analogous tables in earlier pages of the paper (see the printed Table I (10), Table V (10), Table X (6), &c.) which were given exhaustively.]

[Figure: A very wide numerical table (print p. 435) headed by row 147 of the column headings 1^32^4 , 2^24 , 1^32^4 , 12^4 , 1^32^3 , 1^33^2 , 2^23^2 , 13^23 , 12^23 , 1^323 , &c., showing the coefficients of the symmetric functions of weight 10 in the partition rows (10), (91), (82), ..., (1^{10}) ; entries range from +1 through +1814400, and the last row shows totals such as 151200, 302400, 604800, 113400, 226800, 453600, 907200, 1814400, 3628

[Figure: A second wide numerical table (print p. 436), headed by the partitions of weight 10 taken in a different order; this is the table II of the same group, with the rows similarly indexed by partitions (10), (91), (82), ..., (1^{10}) and the columns by monomials of degree 10 in the coefficients; the entries are signed integers including such large values as +225200 and +604800.]

[Figure: A third wide numerical table (print p. 437), headed X (6). **Runs on to p. 439.** The first column heads are 10, 19, 28, 1^28 , 37, 127, 1^27 , 46, 136, 2^26 , 1^226 , 1^26 , 5^2 , 145, ... and the rows are again indexed by the partitions of weight 6 and below. Entries are small signed integers (typically ± 1 through ± 60) and a few are marked with “...” indicating zero.]

[Figure: A fourth wide numerical table (print p. 438), a continuation of Table X (6), with similar column heads 235, 1^235 , 12^25 , 1^225 , 1^45 , 24^2 , 1^24^2 , 3^24 , 1234, 1^334 , 2^34 , 1^22^24 , 1^24 , 1^54 , &c., and the rows the partitions of weight 6.]

[Figure: A fifth wide table (print p. 439), the final part of Table X (6); column heads b^4d^2 , bc^4d , b^2c^3d , ..., b^6c , ... and the rows once more indexed by the partitions of weight 6.]

[End of Paper 147. The detailed numerical entries are extensive; they may be consulted in the original scan.]

148.

MEMOIR ON THE RESULTANT OF A SYSTEM OF TWO EQUATIONS.

[From the Philosophical Transactions of the Royal Society of London, vol. CXLVII for the year 1857, pp. 703–715. Received December 18, 1856,—Read January 8, 1857.]

The Resultant of two equations such as

$$(a, b, \dots | x, y)^m = 0,$$

$$(p, q, \dots | x, y)^n = 0,$$

is, it is well known, a function homogeneous in regard to the coefficients of each equation separately, viz. of the degree n in regard to the coefficients $(a, b, \dots)$ of the first equation, and of the degree m in regard to the coefficients $(p, q, \dots)$ of the second equation; and it is natural to develop the resultant in the form $kAP + k'A'P' + \&c.$, where $A, A', \&c.$ are the combinations (powers and products) of the degree n in the coefficients $(a, b, \dots)$, $P, P', \&c.$ are the combinations of the degree m in the coefficients $(p, q, \dots)$, and $k, k', \&c.$ are mere numerical coefficients. The object of the present memoir is to show how this may be

conveniently effected, either by the method of symmetric functions, or from the known expression of the Resultant in the form of a determinant, and to exhibit the developed expressions for the resultant of two equations, the degrees of which do not exceed 4. With respect to the first method, the formula in its best form, or nearly so, is given in the Algebra of Meyer Hirsch, [for proper title see p. 417], and the application of it is very easy when the necessary tables are calculated: as to this, see my “Memoir on the Symmetric Functions of the Roots of an Equation”¹. But when the expression for the Resultant of two equations is to be calculated without the assistance of such tables, it is I think by far the most simple process to develop the determinant according to the second of the two methods.

Consider first the method of symmetric functions, and to fix the ideas, let the two equations be

$$(a, b, c, d \mid x, y)^3 = 0,$$

$$(p, q, r \mid x, y)^2 = 0.$$

Then writing

$$(a, b, c, d \mid 1, z)^3 = a(1 - \alpha z)(1 - \beta z)(1 - \gamma z),$$

so that

$$\begin{aligned} -\frac{b}{a} &= \alpha + \beta + \gamma = (1), \\ +\frac{c}{a} &= \alpha\beta + \alpha\gamma + \beta\gamma = (1^2), \\ -\frac{d}{a} &= \alpha\beta\gamma = (1^3), \end{aligned}$$

the Resultant is

$$(p, q, r \mid \alpha, 1)^2 \cdot (p, q, r \mid \beta, 1)^2 \cdot (p, q, r \mid \gamma, 1)^2,$$

which is equal to

$$p^3r^3 + q^2r^2(\alpha + \beta + \gamma) + pr^2(\alpha^2 + \beta^2 + \gamma^2) + pqr(\alpha^2\beta + \alpha^2\gamma + \beta^2\alpha + \beta^2\gamma + \gamma^2\alpha + \gamma^2\beta) + \&c.;$$

or adopting the notation for symmetric functions used in the memoir above referred to, this is

$$\left\{ \begin{array}{l} +q^2r^2 \cdot (1) \\ +pr^2 \cdot (2) \\ +q^2r \cdot (1^2) \\ +pqr \cdot (21) \\ +p^2r \cdot (1^3) \\ +q^2 \cdot (2^2) \\ +pq^2 \cdot (21^2) \\ +p^2q \cdot (2^21) \\ +p^3 \cdot (2^3) \end{array} \right.$$

the law of which is best seen by dividing by r^3 and then writing

$$\frac{q}{r} = [1], \quad \frac{p}{r} = [2],$$

and similarly,

$$\frac{pq}{r^2} = [21], \quad \&c.;$$

the expression would then become

$$+1 \cdot 1 + 2 + 1^2 + 21 + 1^3 + 2^2 + 21^2 + 2^21 + 2^3,$$

where the terms within the $[]$ and $()$ are simply all the partitions of the numbers 1, 2, 3, 4, 5, 6, the greatest part being 2, and the greatest number of parts being 3. And in like manner in the general case we have all

¹Philosophical Transactions, 1857, pp. 489–497, [147].

the partitions of the numbers $1, 2, 3, \dots, mn$, the greatest part being n , and the greatest number of parts being m .

The symmetric functions $(1), (2), (1^2)$, &c. are given in the Tables (6) of the Memoir on Symmetric Functions, but it is necessary to remark that in the Tables the first coefficient a is put equal to unity, and consequently that there is a power of the coefficient a to be restored as a factor: this is at once effected by the condition of homogeneity. And it is not by any means necessary to write down (as for clearness of explanation has been done) the preceding expression for the Resultant any portion of it may be taken out directly from one of the Tables (6). For instance, the bracketed portion

$$\begin{aligned} &+ pqr \cdot (21), \\ &+ q^3 \cdot (1^3), \end{aligned}$$

which corresponds to the partitions of the number 3, is to be taken out of the Table III (6) as follows: a portion of this Table (consisting as it happens of consecutive lines and columns, but this is not in general the case) is

$$\begin{array}{c|cc} & = d & bc \\ \hline (21) & +3 & -1 \\ (1^3) & -1 & . \end{array}$$

if in this we omit the sign $=$, and in the outside line write for homogeneity ad instead of d , and in the outside column, first substituting q, p for $1, 2$, then write for homogeneity pqr instead of pq , we have

$$\begin{array}{c|cc} & ad & bc \\ \hline pqr & +3 & -1 \\ q^3 & -1 & . \end{array}$$

viz. $pqr \times (+3ad - 1bc) + q^3 \cdot (-1ad)$, for the value of the portion in question; this is equivalent to

$$\begin{array}{c|cc} & ad & bc \\ \hline pqr & +3 & -1 \\ q^3 & -1 & . \end{array} \quad \text{or, as it may be more conveniently written,}$$

in which form it constitutes a part of the expression given in the sequel for the Resultant of the two functions in question; and similarly the remainder of the expression is at once derived from the Tables (6) I. to VI.

As a specimen of a mode of verification, it may be remarked that the Resultant qua invariant ought, when operated upon by the sum of the two operations,

$$a\partial_b + 2b\partial_c + 3c\partial_d \quad \text{and} \quad p\partial_q + 2q\partial_r,$$

to give a result zero. The results of the two operations are originally obtained in the forms in the first and second columns, and the first column, and the second column, with all the signs reversed, are respectively equal to the third column, and consequently the sum of the first and second columns vanishes, as it ought to do.

Next to explain the second method, viz. the calculation of the resultant from the expression in the form of a determinant. Taking the same example as before, the resultant is

$$\begin{vmatrix} a & b & c & d & . & . & . \\ . & a & b & c & d & . & . \\ . & . & a & b & c & d & . \\ p & q & r & . & . & . & . \\ . & p & q & r & . & . & . \\ . & . & p & q & r & . & . \\ . & . & . & p & q & r & . \end{vmatrix}$$

which may be developed in the form

$$\begin{aligned}
 &+ 12 \cdot 345 \\
 &- 13 \cdot 245 \\
 &+ 14 \cdot 235 \\
 &+ 23 \cdot 145 \\
 &- 15 \cdot 234 \\
 &- 24 \cdot 135 \\
 &+ 25 \cdot 134 \\
 &+ 34 \cdot 125 \\
 &- 35 \cdot 124 \\
 &+ 45 \cdot 123
 \end{aligned}$$

where 12, 13, &c. are the terms of

$$\begin{pmatrix} a, & b, & c, & d \\ a, & b, & c, & d \end{pmatrix}$$

and 123, &c. are the terms of

$$\begin{pmatrix} p, & q, & r \\ p, & q, & r \\ p, & q, & r \end{pmatrix},$$

viz. 12 is the determinant formed with the first and second columns of the upper matrix, 123 is the determinant formed with the first, second and third columns of the lower matrix, and in like manner for the analogous symbols. These determinants must be first calculated, and the remainder of the calculation may then be arranged as follows.

[Figure: A worked-out arithmetic display (print p. 445) giving the array of two-by-two and three-by-three sub-determinants of the upper and lower matrices, together with the rule for combining a “dexter” line of the first square with a “sinister” line of the second to form the entries of a third (product) square.]

where it is to be observed that the figures in the squares of the third column are obtained from those in the corresponding squares of the first and second columns by the ordinary rule for the multiplication of determinants, taking care to multiply the *dexter* lines (i.e. lines in the descending diagonal direction) of the first square by the *sinister* lines (i.e. lines in the direction /) of the second square in order to obtain the sinister lines of the third square. Thus, for instance, the figures in the square

$$\begin{pmatrix} +3 & -1 \\ -1 & 0 \end{pmatrix}$$

are obtained as follows, viz. the first sinister line (+3, −1) by

$$(-1, +1)(-2, +1) = 2 + 1 = +3,$$

$$(-1, +1)(+1, 0) = -1 + 0 = -1,$$

and the second sinister line (−1, 0) by

$$(0, -1)(-2, +1) = 0 - 1 = -1,$$

$$(0, -1)(+1, 0) = +0 - 0 = 0.$$

I have calculated the determinants required for the calculation, by the preceding process, of the Resultant of two quartic equations, and have indeed used them for the verification of the expression as found by the method of symmetric functions; as the determinants in question are useful for other purposes, I think the values are worth preserving.

Table of the Determinants of the Matrices,

$$\begin{pmatrix} a, & b, & c, & d, & e \\ a, & b, & c, & d, & e \\ a, & b, & c, & d, & e \\ a, & b, & c, & d, & e \end{pmatrix}$$

and

$$\begin{pmatrix} p, & q, & r, & s, & t \\ p, & q, & r, & s, & t \\ p, & q, & r, & s, & t \\ p, & q, & r, & s, & t \end{pmatrix}$$

arranged in the form adapted for the calculation of the Resultant of the two quartic equations $(a, b, c, d, e \mid x, y)^4 = 0$, and $(p, q, r, s, t \mid x, y)^4 = 0$, viz.

[Figure: A large foldout-style arrangement (print pp. 446–448) of the sub-determinants of the two coefficient matrices, presented as a diamond-cross pattern of small square sub-blocks of two-by-two, three-by-three and four-by-four determinant values. The arrangement is the standard quartic-quartic resultant tableau on which the subsequent computations are performed; column-sums verify each block. The full set of values, with their signs, is preserved in the original scan.]

The Tables of the Resultants of two equations which I have calculated are as follows.

[Figure: A row of three diamond-cross tableaus (print p. 450) labelled **Table (2, 2)**, **Table (3, 2)**, **Table (4, 2)**, giving the Resultant of $(a, b, c \mid x, y)^2$ with $(p, q, r \mid x, y)^2$; of $(a, b, c, d \mid x, y)^3$ with $(p, q, r \mid x, y)^2$; and of $(a, b, c, d, e \mid x, y)^4$ with $(p, q, r \mid x, y)^2$ respectively.]

[Figure: A row of two further diamond-cross tableaus (print p. 451) labelled **Table (3, 3)** and **Table (4, 3)**, giving the Resultant of $(a, b, c, d \mid x, y)^3$ with $(p, q, r, s \mid x, y)^3$, and of $(a, b, c, d, e \mid x, y)^4$ with $(p, q, r, s \mid x, y)^3$ respectively.]

[Figure: A larger diamond-cross tableau (print p. 452) labelled **Table (4, 4)**, giving the Resultant of $(a, b, c, d, e \mid x, y)^4$ with $(p, q, r, s, t \mid x, y)^4$.]

[End of paper 148.]

149.

ON THE SYMMETRIC FUNCTIONS OF THE ROOTS OF CERTAIN SYSTEMS OF TWO EQUATIONS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLVII for the year 1857, pp. 717–726. Received December 18, 1856,—Read January 8, 1857.]

Suppose in general that $\phi = 0$, $\psi = 0$, &c. denote a system of $(n - 1)$ equations between the n variables $(x, y, z, \dots)$, where the functions ϕ, ψ , &c. are quantics (i.e. rational and integral homogeneous functions) of the variables. Any values $(x_1, y_1, z_1, \dots)$ satisfying the equations, are said to constitute a set of roots of the system; the roots of the same set are, it is clear, only determinate to a common factor *près*, i.e. only the ratios *inter se* and not the absolute magnitudes of the roots of a set are determinate. The number of sets, or the degree of the system, is equal to the product of the degrees of the component equations. Imagine a function of the roots which remains unaltered when any two sets $(x_1, y_1, z_1, \dots)$ and $(x_2, y_2, z_2, \dots)$ are interchanged (that is, when x_1 and x_2 , y_1 and y_2 , &c. are simultaneously interchanged), and which is besides homogeneous of the same degree as regards each entire set of roots, although not of necessity homogeneous as regards the different roots of the same set; thus, for example, if the sets are $(x_1, y_1), (x_2, y_2)$, then the functions $x_1x_2, x_1y_2 + x_2y_1, y_1y_2$ are each of them of the form in question; but the first and third of these functions, although homogeneous of the first degree in regard to each entire set, are not homogeneous as regards the two variables of each set. A function of the above-mentioned form may, for shortness, be termed a symmetric function of the roots; such function (disregarding an arbitrary factor depending on the common factors which enter implicitly into the different sets of roots) will be a rational and integral function of the coefficients of the equations, i.e. any symmetric function of the roots may be considered as a rational and integral function of the coefficients. The general process for the investigation of such expression for a symmetric function of the roots is indicated in Professor Schläfli's Memoir, "Ueber die Resultante eines Systemes mehrerer algebraischer Gleichungen," *Vienna Transactions*, t. IV. (1852). The process is as follows:—Suppose that we know the resultant of a system of equations, one or more of them being linear; then if $\phi = 0$ be the linear equation or one of the linear equations of the system, the resultant will be of the form $\phi_1\phi_2\dots$, where ϕ_1, ϕ_2 , &c. are what the function ϕ becomes upon substituting therein the different sets $(x_1, y_1, z_1, \dots), (x_2, y_2, z_2, \dots)$ of the remaining $(n - 1)$ equations

$\psi = 0$, $\chi = 0$, &c.; comparing such expression with the given value of the resultant, we have expressed in terms of the coefficients of the functions ψ, χ , &c., certain symmetric functions which may be called the *fundamental* symmetric functions of the roots of the system $\psi = 0$, $\chi = 0$, &c.; these are in fact the symmetric functions of the first degree in respect to each set of roots. By the aid of these fundamental symmetric functions, the other symmetric functions of the roots of the system $\psi = 0$, $\chi = 0$, &c. may be expressed in terms of the coefficients, and then combining with these equations a non-linear equation $\Phi = 0$, the resultant of the system $\Phi = 0$, $\psi = 0$, $\chi = 0$, &c. will be what the function $\Phi_1\Phi_2\dots$ becomes, upon substituting therein for the different symmetric functions of the roots of the system $\psi = 0$, $\chi = 0$, &c. the expressions for these functions in terms of the coefficients. We thus pass from the resultant of a system $\phi = 0$, $\psi = 0$, $\chi = 0$, &c., to that of a system $\Phi = 0$, $\psi = 0$, $\chi = 0$, &c., in which the linear function ϕ is replaced by the non-linear function Φ . By what has preceded, the symmetric functions of the roots of a system of $(n - 1)$ equations depend on the resultant of the system obtained by combining the $(n - 1)$ equations with an arbitrary linear equation; and moreover, the resultant of any system of n equations depends ultimately upon the resultant of a system of the same number of equations, all except one being linear; but in this case the linear equations determine the ratios of the variables or (disregarding a common factor) the values of the variables, and by substituting these values in the remaining equation we have the resultant of the system. The process leads, therefore, to the expressions for the symmetric functions of the roots of any system of $(n - 1)$ equations, and also to the expression for the resultant of any system of n equations. Professor Schläfli discusses in the general case the problem of showing how the expressions for the fundamental symmetric functions lead to those of the other symmetric functions, but it is not necessary to speak further of this portion of his investigations. The object of the present Memoir is to apply the process to two particular cases, viz. I propose to obtain thereby the expressions for the simplest symmetric functions (after the fundamental ones) of the following systems of two ternary equations; that is, first, a linear equation and a quadric equation; and secondly, a linear equation and a cubic equation.

First, consider the two equations

$$\begin{aligned} (\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h} \mid x, y, z)^2 &= 0, \\ (\alpha, \beta, \gamma \mid x, y, z) &= 0, \end{aligned}$$

and join to these the arbitrary linear equation

$$(\xi, \eta, \zeta \mid x, y, z) = 0.$$

then the two linear equations give $x : y : z$ in terms of (α, β, γ) and (ξ, η, ζ) , and substituting in the quadratic equation, we have for the resultant of the three equations,

$$\square^{-1} \cdot (\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h} \mid \beta\zeta - \gamma\eta, \gamma\xi - \alpha\zeta, \alpha\eta - \beta\xi)^2 = 0,$$

which may be represented by

$$\square^{-1} \cdot (\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h} \mid \xi, \eta, \zeta)^2 = 0,$$

where the coefficients are given by means of the Table.

[Table (print p. 456): a six-by-six array giving each of $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h}$ as a linear combination of $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h}$ with coefficients quadratic in α, β, γ .]

	$\mathbf{a}$	$\mathbf{b}$	$\mathbf{c}$	$\mathbf{f}$	$\mathbf{g}$	$\mathbf{h}$
$\mathbf{a} =$	.	$+\gamma^2$	$+\beta^2$	$-2\beta\gamma$	.	.
$\mathbf{b} =$	$+\gamma^2$	.	$+\alpha^2$	.	$-2\gamma\alpha$	.
$\mathbf{c} =$	$+\beta^2$	$+\alpha^2$	.	.	.	$-2\alpha\beta$
$\mathbf{f} =$	$-\beta\gamma$	$-\gamma\alpha$	.	$-\alpha^2$	$+\alpha\beta$	$+\gamma\alpha$
				$+\alpha\beta$		$+\beta\gamma$
$\mathbf{g} =$	.	.	$-\alpha\beta$	$+\gamma\alpha$	$-\beta^2$	.
					$+\gamma\beta$	
$\mathbf{h} =$	.	.	.	.	$+\beta\gamma$	.

viz. $\mathbf{a} = \mathbf{b}\gamma^2 + \mathbf{c}\beta^2 - 2\mathbf{f}\beta\gamma$, &c.

But if the roots of the given system are

$$(x_1, y_1, z_1), (x_2, y_2, z_2),$$

then the resultant of the three equations will be

$$(\xi, \eta, \zeta \mid x_1, y_1, z_1)(\xi, \eta, \zeta \mid x_2, y_2, z_2),$$

and comparing the two expressions, we have

$$\begin{aligned} a &= x_1 x_2, \\ b &= y_1 y_2, \\ c &= z_1 z_2, \\ 2f &= y_1 z_2 + y_2 z_1, \\ 2g &= z_1 x_2 + z_2 x_1, \\ 2h &= x_1 y_2 + x_2 y_1, \end{aligned}$$

which are the expressions for the six fundamental symmetric functions, or symmetric functions of the first degree in each set, of the roots of the given system.

By forming the powers and products of the second order $a^2, ab, \&c.$, we obtain linear relations between the symmetric functions of the second degree in respect to each set of roots. The number of equations is precisely equal to that of the symmetric functions of the form in question, and the solution of the linear equations gives

$$\begin{aligned} a^2 &= x_1^2 x_2^2, \\ ab &= x_1 y_1 x_2 y_2, \\ 4f^2 - 2bc &= y_1^2 z_2^2 + y_2^2 z_1^2, \\ 4g^2 - 2ca &= z_1^2 x_2^2 + z_2^2 x_1^2, \\ 4h^2 - 2ab &= x_1^2 y_2^2 + x_2^2 y_1^2, \\ 2af &= x_1 y_1 z_1 x_2 + x_2 y_2 z_2 x_1, \\ 2bg &= y_1 z_1 x_1 y_2 + y_2 z_2 x_2 y_1, \\ 2ch &= z_1 x_1 y_1 z_2 + z_2 x_2 y_2 z_1, \\ 4gh - 2af &= x_1^2 y_2 z_2 + x_2^2 y_1 z_1, \\ 4hf - 2bg &= y_1^2 z_2 x_2 + y_2^2 z_1 x_1, \\ 4fg - 2ch &= z_1^2 x_2 y_2 + z_2^2 x_1 y_1, \\ 2bf &= y_1^2 y_2 z_2 + y_2^2 y_1 z_1, \\ 2cg &= z_1^2 z_2 x_2 + z_2^2 z_1 x_1, \\ 2ah &= x_1^2 x_2 y_2 + x_2^2 x_1 y_1, \\ 2cf &= z_1^2 y_2 z_2 + z_2^2 y_1 z_1, \\ 2bh &= y_1^2 x_2 y_2 + y_2^2 x_1 y_1. \end{aligned}$$

Proceeding next to the powers and products of the third order $a^3, a^2b, \&c.$, the total number of linear relations between the symmetric functions of the third degree in respect to each set of roots exceeds by unity the number of the symmetric functions of the form in question; in fact the expressions for $abc, af^2, bg^2, ch^2, fgh,$

contain, not five, but only four symmetric functions of the roots; for we have

$$\begin{aligned}
 abc &= x_1 y_1 z_1 \cdot x_2 y_2 z_2, \\
 4af^2 &= \{x_1^2 y_2^2 z_2^2 + x_2^2 y_1^2 z_1^2\} + 2x_1 y_1 z_1 x_2 y_2 z_2, \\
 4bg^2 &= \{y_1^2 z_2^2 x_2^2 + y_2^2 z_1^2 x_1^2\} + 2x_1 y_1 z_1 x_2 y_2 z_2, \\
 4ch^2 &= \{z_1^2 x_2^2 y_2^2 + z_2^2 x_1^2 y_1^2\} + 2x_1 y_1 z_1 x_2 y_2 z_2, \\
 8fgh &= \{x_1^2 y_2^2 x_2 z_2 + x_2^2 y_1^2 x_1 z_1\} \\
 &\quad + \{y_1^2 z_2^2 y_2 x_2 + y_2^2 z_1^2 y_1 x_1\} \\
 &\quad + \{z_1^2 x_2^2 z_2 y_2 + z_2^2 x_1^2 z_1 y_1\} \\
 &\quad + 2x_1 y_1 z_1 x_2 y_2 z_2,
 \end{aligned}$$

and consequently the quantities a, b, c, f, g, h are not independent, but are connected by the equation

$$abc - af^2 - bg^2 - ch^2 + 2fgh = 0,$$

an equation, which is in fact verified by the foregoing values of a, &c. in terms of the coefficients of the given system.

The expressions for the symmetric functions of the third degree considered as functions of a, b, c, f, g, h, are consequently not absolutely determinate, but they may be modified by the addition of the term $\lambda(abc - af^2 - bg^2 - ch^2 + 2fgh)$, where λ is an indeterminate numerical coefficient.

The simplest expressions are those obtained by disregarding the preceding equation for fgh, and the entire

system then becomes

$$\begin{aligned}
 a^3 &= x_1^3 x_2^3, \\
 b^3 &= y_1^3 y_2^3, \\
 c^3 &= z_1^3 z_2^3, \\
 a^2 b &= x_1^2 y_1 x_2^2 y_2, \\
 bc^2 &= y_1 z_1^2 y_2 z_2^2, \\
 ab^2 &= x_1 y_1^2 x_2 y_2^2, \\
 abc &= x_1 y_1 z_1 x_2 y_2 z_2, \\
 2a^2 f &= x_1^2 y_2 z_2 x_2 + x_2^2 y_1 z_1 x_1, \\
 2b^2 g &= y_1^2 z_2 x_2 y_2 + y_2^2 z_1 x_1 y_1, \\
 2c^2 h &= z_1^2 x_2 y_2 z_2 + z_2^2 x_1 y_1 z_1, \\
 2a^2 g &= x_1^2 z_2 x_2^2 + x_2^2 z_1 x_1^2, \\
 2b^2 h &= y_1^2 x_2 y_2^2 + y_2^2 x_1 y_1^2, \\
 2c^2 f &= z_1^2 y_2 z_2^2 + z_2^2 y_1 z_1^2, \\
 2a^2 h &= x_1^2 x_2^2 y_2 + x_2^2 x_1^2 y_1, \\
 2b^2 f &= y_1^2 y_2^2 z_2 + y_2^2 y_1^2 z_1, \\
 2c^2 g &= z_1^2 z_2^2 x_2 + z_2^2 z_1^2 x_1, \\
 2bcf &= y_1 z_1 y_2^2 z_2^2 + y_2 z_2 y_1^2 z_1^2, \\
 2cag &= z_1 x_1 z_2^2 x_2^2 + z_2 x_2 z_1^2 x_1^2, \\
 2abh &= x_1 y_1 x_2^2 y_2^2 + x_2 y_2 x_1^2 y_1^2, \\
 2bch &= y_1 z_1 x_2 y_2 z_2^2 + y_2 z_2 x_1 y_1 z_1^2, \\
 2caf &= z_1 x_1 x_2 y_2 z_2^2 + z_2 x_2 x_1 y_1 z_1^2, \\
 2abg &= x_1 y_1 z_2 x_2 y_2^2 + x_2 y_2 z_1 x_1 y_1^2, \\
 4af^2 - 2abc &= x_1 y_1 z_1 x_2^3 + x_2 y_2 z_2 x_1^3, \\
 4bg^2 - 2abc &= y_1 z_1 x_1 y_2^3 + y_2 z_2 x_2 y_1^3, \\
 4ch^2 - 2abc &= z_1 x_1 y_1 z_2^3 + z_2 x_2 y_2 z_1^3, \\
 4bf^2 - 2b^2 c &= y_1^2 z_2^2 y_2 + y_2^2 z_1^2 y_1, \\
 4cg^2 - 2c^2 a &= z_1^2 x_2^2 z_2 + z_2^2 x_1^2 z_1, \\
 4ah^2 - 2a^2 b &= x_1^2 y_2^2 x_2 + x_2^2 y_1^2 x_1, \\
 4cf^2 - 2bc^2 &= z_1^2 y_2 z_2^2 + z_2^2 y_1 z_1^2, \\
 4ag^2 - 2ca^2 &= x_1^2 z_2 x_2^2 + x_2^2 z_1 x_1^2, \\
 4bh^2 - 2ab^2 &= y_1^2 x_2 y_2^2 + y_2^2 x_1 y_1^2, \\
 4agh - 2a^2 f &= x_1^2 x_2 y_2 z_2 + x_2^2 x_1 y_1 z_1, \\
 4bhf - 2b^2 g &= y_1^2 x_2 y_2 z_2 + y_2^2 x_1 y_1 z_1, \\
 4cfg - 2c^2 h &= z_1^2 x_2 y_2 z_2 + z_2^2 x_1 y_1 z_1, \\
 4bgh - 2abf &= y_1 z_1 x_2 y_2 z_2 + y_2 z_2 x_1 y_1 z_1, \\
 4chf - 2bcg &= z_1 x_1 y_2 z_2^2 + z_2 x_2 y_1 z_1^2, \\
 4afg - 2cah &= x_1 y_1 z_2 x_2^2 + x_2 y_2 z_1 x_1^2, \\
 4cgh - 2acf &= y_1 z_1 x_2 z_2^2 + y_2 z_2 x_1 z_1^2, \\
 4ahf - 2bag &= z_1 x_1 y_2 x_2^2 + z_2 x_2 y_1 x_1^2, \\
 4bfg - 2cbh &= x_1 y_1^2 z_2^2 y_2 + x_2 y_2^2 z_1^2 y_1, \\
 8f^2 g - 4chf - 2bcg &= z_1^2 x_2 y_2^2 + z_2^2 x_1 y_1^2, \\
 8g^2 h - 4afg - 2cah &= x_1^2 y_2 z_2^2 + x_2^2 y_1 z_1^2, \\
 8h^2 f - 4bgh - 2abf &= y_1^2 z_2 x_2^2 + y_2^2 z_1 x_1^2, \\
 8fg^2 - 4chg - 2acf &= z_1^2 x_2 y_2^2 + z_2^2 x_1 y_1^2,
 \end{aligned}$$

Secondly, consider the system of equations

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l} \mid x, y, z)^3 = 0,$$

$$(\alpha, \beta, \gamma \mid x, y, z) = 0,$$

where the cubic function written at full length is

$$\mathfrak{a}x^3 + \mathfrak{b}y^3 + \mathfrak{c}z^3 + 3\mathfrak{f}y^2z + 3\mathfrak{g}z^2x + 3\mathfrak{h}x^2y + 3\mathfrak{i}yz^2 + 3\mathfrak{j}zx^2 + 3\mathfrak{k}xy^2 + 6\mathfrak{l}xyz.$$

Joining to the system the linear equation

$$(\xi, \eta, \zeta \mid x, y, z) = 0,$$

the linear equations give

$$x : y : z = \beta\zeta - \gamma\eta : \gamma\xi - \alpha\zeta : \alpha\eta - \beta\xi,$$

and the resultant is

$$\square^{-1} \cdot (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l} \mid \beta\zeta - \gamma\eta, \gamma\xi - \alpha\zeta, \alpha\eta - \beta\xi)^3 = 0,$$

which may be represented by

$$\square^{-1} \cdot (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l} \mid \xi, \eta, \zeta)^3 = 0,$$

where the coefficients $\mathfrak{a}, \mathfrak{b}$, &c. are given by means of the Table.

[Table (print p. 460): a ten-by-ten array giving each of $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l}$ as a linear combination of $\mathfrak{a}, \dots, \mathfrak{l}$ with coefficients cubic in α, β, γ . The leading entries are $\mathfrak{a} = -3\mathfrak{f}\beta\gamma + 3\mathfrak{i}\beta^2\gamma + \dots$, $\mathfrak{b} = -\gamma^3\mathfrak{a} + \alpha^3\mathfrak{c} - 3\gamma\alpha^2\mathfrak{g} + 3\gamma^2\alpha\mathfrak{j} + \dots$, &c.; the full pattern of signs and the higher-order terms in $\beta\gamma, \gamma\alpha, \alpha\beta$ are preserved in the original scan.]

viz. $\mathfrak{a} = -\mathfrak{c}\beta^3 + \mathfrak{b}\gamma^3 - 3\mathfrak{f}\beta\gamma^2 - 3\mathfrak{i}\beta^2\gamma$, &c.

But if the roots of the given system are

$$(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3),$$

then the resultant of the three equations may also be represented by

$$(\xi, \eta, \zeta \mid x_1, y_1, z_1)(\xi, \eta, \zeta \mid x_2, y_2, z_2)(\xi, \eta, \zeta \mid x_3, y_3, z_3),$$

and comparing with the former expression, we find:

$$\begin{aligned} \mathfrak{a} &= x_1x_2x_3, \\ \mathfrak{b} &= y_1y_2y_3, \\ \mathfrak{c} &= z_1z_2z_3, \\ 3\mathfrak{f} &= y_1y_2z_3 + y_2y_3z_1 + y_3y_1z_2, \\ 3\mathfrak{h} &= x_1x_2y_3 + x_2x_3y_1 + x_3x_1y_2, \\ 3\mathfrak{i} &= y_1z_2z_3 + y_2z_3z_1 + y_3z_1z_2, \\ 3\mathfrak{k} &= x_1y_2y_3 + x_2y_3y_1 + x_3y_1y_2, \\ 6\mathfrak{l} &= x_1y_2z_3 + x_1y_3z_2 + x_2y_1z_3 + x_2y_3z_1 + x_3y_1z_2 + x_3y_2z_1. \end{aligned}$$

But there is in the present case a relation independent of the quantities $\mathfrak{a}$, &c., viz. we have $(\alpha, \beta, \gamma \mid x_1, y_1, z_1) = 0$, $(\alpha, \beta, \gamma \mid x_2, y_2, z_2) = 0$, $(\alpha, \beta, \gamma \mid x_3, y_3, z_3) = 0$, and thence eliminating the coefficients (α, β, γ) , we find

$$\nabla = x_1y_2z_3 + x_2y_3z_1 + x_3y_1z_2 - x_1y_3z_2 - x_2y_1z_3 - x_3y_2z_1 = 0.$$

By forming the powers and products of the second degree $\mathfrak{a}^2, \mathfrak{a}\mathfrak{b}$, &c., we obtain 55 equations between the symmetric functions of the second degree in each set of roots. But we have $\nabla^2 =$ a symmetric function

of the roots, and thus the entire number of linear relations is 56, and this is in fact the number of the symmetric functions of the second degree in each set. I use for shortness the sign S to denote the sum of the distinct terms obtained by permuting the different sets of roots, so that the equations for the fundamental symmetric functions are

$$a = x_1x_2x_3,$$

$$b = y_1y_2y_3,$$

$$c = z_1z_2z_3,$$

$$3f = Sy_1y_2z_3,$$

$$3g = Sz_1z_2x_3,$$

$$3h = Sx_1x_2y_3,$$

$$3i = Sy_1z_2z_3,$$

$$3j = Sz_1x_2x_3,$$

$$3k = Sx_1y_2y_3,$$

$$6l = Sx_1y_2z_3;$$

then the complete system of expressions for the symmetric functions of the second order is as follows, viz.

$$\begin{aligned}
a^2 &= x_1^2 x_2^2 x_3^2, \\
b^2 &= y_1^2 y_2^2 y_3^2, \\
c^2 &= z_1^2 z_2^2 z_3^2, \\
bc &= y_1 z_1 y_2 z_2 y_3 z_3, \\
ca &= z_1 x_1 z_2 x_2 z_3 x_3, \\
ab &= x_1 y_1 x_2 y_2 x_3 y_3, \\
3af &= S x_1 y_1 x_2 y_2 x_3, \\
3ch &= S z_1 x_1 z_2 x_2 y_3 z_3, \\
3bf &= S y_1^2 y_2^2 y_3 z_3, \\
3cg &= S z_1^2 z_2 z_3 x_3, \\
3ah &= S x_1^2 x_2 x_3 y_3, \\
3cf &= S y_1 z_1 y_2 z_2 z_3^2, \\
3ag &= S z_1 x_1 z_2 x_3^2, \\
3bh &= S x_1 y_1 x_2 y_3^2, \\
3ai &= S x_1 y_1 z_2 x_2 x_3 z_3, \\
3bj &= S y_1 z_1 x_2 y_2 y_3 x_3, \\
3ck &= S z_1 x_1 y_2 z_2 z_3 x_3, \\
3bi &= S y_1^2 y_2 z_2 y_3 z_3, \\
3cj &= S z_1^2 z_2 x_2 z_3 x_3, \\
3ak &= S x_1^2 x_2 y_2 x_3 y_3, \\
3ci &= S y_1 z_1 z_2 x_2 z_3^2, \\
3aj &= S z_1 x_1 x_2 x_3^2, \\
3bk &= S x_1 y_1 y_2 y_3^2, \\
6al &= S x_1^2 x_2 y_2 x_3 z_3, \\
6bl &= S y_1^2 y_2 x_2 y_3 z_3, \\
6cl &= S z_1^2 z_2 x_2 y_3 z_3, \\
9f^2 - 6bi &= S y_1^2 y_2^2 z_3^2, \\
9g^2 - 6cj &= S z_1^2 z_2^2 x_3^2, \\
9h^2 - 6ak &= S x_1^2 x_2^2 y_3^2, \\
9i^2 - 6cf &= S y_1^2 z_2^2 z_3^2, \\
9j^2 - 6ag &= S z_1^2 x_2^2 x_3^2, \\
9k^2 - 6bh &= S x_1^2 y_2^2 y_3^2, \\
9fg - 3ck &= S z_1 x_1 y_2 z_2 y_3^2, \\
9gh - 3ai &= S y_1 z_1 z_2 x_2 z_3^2, \\
9hf - 3bj &= S z_1 x_1 x_2 y_2 x_3^2, \\
9jk - 3af &= S x_1^2 x_2 y_2 y_3 z_3, \\
9ki - 3bg &= S y_1^2 y_2 z_2 z_3 x_3, \\
9ij - 3ca &= S z_1^2 z_2 x_2 x_3 y_3, \\
9fi - 3bc &= S y_1^2 y_2 z_2 z_3^2, \\
9gj - 3ca &= S z_1^2 z_2 x_2 x_3^2, \\
9hk - 3ab &= S x_1^2 x_2 y_2 y_3^2, \\
3(2fi + gk + hj - l^2) &= S x_1 y_1 z_1 y_2 z_2 y_3 z_3, \\
3(2gj - kh - hi + l^2) &= S x_1 y_1 z_1 z_2 x_2 z_3 x_3, \\
3(2hk - fi - gk + l^2) &= S x_1 y_1 z_1 x_2 y_2 x_3 y_3,
\end{aligned}$$

As an instance of the application of the formulae, let it be required to eliminate the variables from the three equations,

$$\begin{aligned}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l} \mid x, y, z)^3 &= 0, \\ (\mathfrak{a}', \mathfrak{b}', \mathfrak{c}', \mathfrak{f}', \mathfrak{g}', \mathfrak{h}' \mid x, y, z)^2 &= 0, \\ (\alpha, \beta, \gamma \mid x, y, z) &= 0.\end{aligned}$$

This may be done in two different ways; first, representing the roots of the linear equation and the quadric equation by $(x_1, y_1, z_1), (x_2, y_2, z_2)$, the resultant will be

$$(\mathfrak{a}, \dots \mid x_1, y_1, z_1)^3 \cdot (\mathfrak{a}, \dots \mid x_2, y_2, z_2)^3,$$

which is equal to $\mathfrak{a}^2 \cdot x_1^3 x_2^3 + \&c.$, where the symmetric functions $x_1^3 x_2^3$, &c. are given by the formulae $\mathfrak{a}^2 = x_1^2 x_2^2$, &c., in which, since the coefficients of the quadratic equation are $(\mathfrak{a}', \mathfrak{b}', \mathfrak{c}', \mathfrak{f}', \mathfrak{g}', \mathfrak{h}')$, I have written $\mathfrak{a}'$ instead of $\mathfrak{a}$. Next, if the roots of the linear equation and the cubic equation are represented by $(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)$, then the resultant will be

$$(\mathfrak{a}', \dots \mid x_1, y_1, z_1)^2 \cdot (\mathfrak{a}', \dots \mid x_2, y_2, z_2)^2 \cdot (\mathfrak{a}', \dots \mid x_3, y_3, z_3)^2,$$

which is equal to

$$\mathfrak{a}'^3 \cdot x_1^2 x_2^2 x_3^2 + \&c.,$$

the symmetric functions $x_1^2 x_2^2 x_3^2$, &c. being given by the formulae $\mathfrak{a}^2 = x_1^2 x_2^2 x_3^2$, &c. The expression for the Resultant is in each case of the right degree, viz. of the degrees 6, 3, 2, in the coefficients of the linear, the quadric, and the cubic equations respectively: the two expressions, therefore, can only differ by a numerical factor, which might be determined without difficulty. The third expression for the resultant, viz.

$$\square^{-1} \cdot (\alpha, \beta, \gamma \mid x_1, y_1, z_1) \cdot (\alpha, \beta, \gamma \mid x_2, y_2, z_2) \cdots (\alpha, \beta, \gamma \mid x_6, y_6, z_6),$$

(where $(x_1, y_1, z_1), \dots, (x_6, y_6, z_6)$ are the roots of the cubic and quadratic equations) compared with the foregoing value, leads to expressions for the fundamental symmetric functions of the cubic and quadratic equations, and thence to expressions for the other symmetric functions of these two equations; but it would be difficult to obtain the actually developed values even of the fundamental symmetric functions. I hope to return to the subject, and consider in a general point of view the question of the formation of the expressions for the other symmetric functions by means of the expressions for the fundamental symmetric functions.

[End of paper 149.]